PROJECT DOCUMENTATION

MedTrack DV

Hospital Operations & Patient Analytics Dashboard Suite

Developed in Microsoft Power BI

Prepared By Moulika Rudraraju

INFOSYS SPRINGBOARD INTERNSHIP 7.0

Data Visualization Track — Project Submission

Under the Guidance of

Manisha Panda, Data Analyst

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Contents

Abstract

Hospitals generate large volumes of operational data across admissions, discharges, departments, beds, patient stay, billing, and resource allocation. Without a consolidated analytical view, it becomes difficult to identify trends, compare department performance, monitor patient flow, and evaluate resource utilization.

MedTrack DV is a hospital operations and patient analytics dashboard suite developed in Microsoft Power BI. The project converts patient admission and hospital operational data into interactive visual insights that support monitoring of hospital performance, patient flow, department-level performance, and healthcare resource utilization. The final report contains four interconnected analytical dashboards — Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization — supported by About and Help pages for usability.

The hospital dataset used for the project contains 200 patient records across 9 departments. The data model also includes a department-level resource table that supports staff allocation and equipment utilization analysis. Through KPI cards, trend visuals, department comparisons, slicers, and page navigation, MedTrack DV provides a clear analytical interface for understanding hospital operations and demonstrates an end-to-end data visualization workflow, from data preparation and KPI engineering to interactive dashboard development.

1. Introduction

1.1 Motivation

Hospitals routinely collect operational data covering admissions, discharges, departments, treatment types, bed status, billing, and patient satisfaction. While this data is valuable, it is often spread across disconnected records and spreadsheets, making it difficult for administrators to build a timely, consolidated picture of hospital performance.

Without a unified analytical view, patterns such as rising readmission rates, department-level bottlenecks, or under-utilized resources can go unnoticed until they become significant operational or financial problems. The motivation behind MedTrack DV is to address this gap by transforming raw hospital operational data into an interactive, easy-to-read dashboard suite that surfaces these patterns as they emerge.

1.2 Problem Definition

Hospital operations data is typically fragmented across admissions, discharge, billing, and department-level records, which creates several practical challenges:

- Lack of a single consolidated view of hospital performance and patient flow
- Difficulty comparing departments on volume, length of stay, readmissions, and revenue
- Limited visibility into bed, staff, and equipment utilization
- No easy way to track monthly trends in admissions, discharges, and readmissions
- Manual reporting is slow, error-prone, and not interactive

MedTrack DV addresses these challenges through a unified Power BI dashboard suite that consolidates patient-level and department-level data into a single, interactive analytical model.

1.3 Objectives of the Project

- To design and develop a consolidated hospital operations and patient analytics dashboard suite in Power BI
- To engineer meaningful hospital KPIs covering admissions, occupancy, readmissions, satisfaction, and revenue
- To build a data model that connects patient-level and department-level resource data
- To provide department-wise comparisons of volume, length of stay, readmissions, and revenue
- To monitor bed, staff, and equipment utilization across departments
- To support interactive exploration through slicers, cross-filtering, and page navigation
- To present findings through a clear, professional, and portfolio-ready report

2. Literature Survey

Hospital operations analytics sits at the intersection of business intelligence (BI) tooling and healthcare performance management. Modern BI platforms such as Microsoft Power BI, Tableau, and Qlik are widely used across industries, including healthcare, to convert transactional and operational data into interactive dashboards that support decision-making without requiring end users to write queries.

In a healthcare operations context, published dashboard practice commonly centers on a small set of recurring themes: patient-flow monitoring (admissions, discharges, and length of stay), department-level performance comparison, resource utilization (beds, staff, and equipment), and patient experience indicators such as satisfaction and readmission rate. These themes directly informed the KPI selection and dashboard structure adopted in MedTrack DV.

A recurring observation in BI practice is that dashboards are most useful when they combine summary-level KPI cards with trend visuals and drill-down comparisons, rather than presenting either in isolation. This principle shaped the layout of each MedTrack DV page, which pairs top-level KPI cards with monthly trend charts and department-wise comparison visuals.

Based on this review of common BI and healthcare-analytics practice, a gap exists between simple single-purpose reporting tools and a complete, centralized, interactive solution that connects patient-level and resource-level data in one model. MedTrack DV is designed to address that gap within the scope of an academic and portfolio analytics project.

3. System Analysis

3.1 Existing System

Prior to a consolidated dashboard, hospital operational reporting typically relies on manual or partially automated methods that limit visibility and responsiveness:

- Manual/Spreadsheet Tracking — department or admissions data maintained in separate spreadsheets, which is time-consuming and error-prone
- Siloed Departmental Records — each department tracks its own volume and performance separately, with no cross-department comparison
- Static or Periodic Reports — reports generated infrequently, so trends are noticed late
- No Unified Resource View — bed, staff, and equipment utilization tracked separately from patient data, if at all

Limitations of the existing approach include the absence of a centralized system, delayed visibility into readmission and occupancy trends, limited department-level comparability, and significant manual effort to produce even basic reports.

3.2 Proposed System

The proposed system, MedTrack DV, introduces a centralized Power BI dashboard suite that connects patient-level operational data with department-level resource data in a single model, enabling interactive, real-time-style analysis.

- Centralized Dashboards — Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization in one report
- KPI-Driven Monitoring — DAX measures summarize admissions, occupancy, readmissions, satisfaction, and revenue
- Department Comparison — volume, length of stay, readmission rate, and revenue compared side by side
- Resource Visibility — bed occupancy, staff allocation, and equipment utilization tracked by department
- Interactive Exploration — slicers, cross-filtering, and page navigation for self-service analysis

3.3 Requirement Specification Overview

3.3.1 Purpose

The purpose of this specification is to describe the functional and non-functional requirements of the MedTrack DV dashboard suite — what the report should display, how users interact with it, and the data it depends on.

3.3.2 Scope

The scope covers preparation of hospital admission and department-resource data, KPI engineering, dashboard design across four analytical areas, interactivity (slicers, navigation, cross-filtering), and packaging the result as a Power BI report (.pbix). The scope does not include a live hospital information system, direct clinical decision support, or real-time data feeds — the current version is built on a prepared, static dataset.

3.3.3 Overall Description

MedTrack DV consists of the following major components:

- Data Preparation Layer — cleaning and shaping the hospital and resource datasets in Power Query
- Data Model — a relationship between the patient-level hospital table and the department-level resource table
- KPI / DAX Layer — measures that calculate admissions, occupancy, readmission rate, satisfaction, and revenue
- Dashboard Layer — four report pages plus About and Help pages
- Interactivity Layer — slicers, cross-filtering, and page-navigation buttons

4. Requirement Specifications

4.1 Functional Requirements

- The system must load and model patient-level and department-level resource data
- The system must calculate hospital KPIs (admissions, occupancy, readmission rate, satisfaction, revenue) using DAX measures
- The system must display a consolidated Hospital Overview page with KPI cards and monthly trend visuals
- The system must provide a Patient Flow page showing admissions, discharges, average stay, and peak load
- The system must provide a Department Analytics page comparing departments on volume, stay, readmissions, and revenue
- The system must provide a Resource Utilization page showing bed, staff, and equipment utilization by department
- The system must support slicers and cross-filtering so users can explore the data by department, date, and other dimensions
- The system must provide page navigation, including About and Help pages

4.2 Software Requirements

- Reporting Tool: Microsoft Power BI Desktop
- Data Preparation: Power Query (M language)
- KPI Calculation: DAX (Data Analysis Expressions)
- Data Sources: CSV / structured hospital and resource datasets
- Operating System: Windows 10/11

4.3 Hardware Requirements

- Processor: Intel i3 or higher (i5 recommended for smoother report rendering)
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: At least 2 GB free disk space for Power BI Desktop and project files
- Display: Standard monitor with at least 1366×768 resolution for comfortable dashboard viewing
- Network: Internet connection for installation, updates, and optional publishing to Power BI Service

5. System Design

5.1 Introduction of Technologies Used

Microsoft Power BI Desktop — used to build the full report: data import, modeling, DAX measures, and visual design. Its drag-and-drop visual editor and native DAX engine make it well suited to KPI-driven operational dashboards.

Power Query — used during data preparation to clean, standardize, and shape the hospital and resource datasets (date formatting, consistent department labels, numeric field preparation) before loading them into the model.

DAX (Data Analysis Expressions) — used to define the KPI measures that drive every KPI card and chart in the report, including admissions, occupancy, readmission rate, satisfaction, and revenue.

Data Model (Star-style relationship) — the hospital (patient-level) table and the department-level resource table are related through the Department field, enabling department-based resource analysis alongside patient-level metrics.

5.2 Data Sources and Preparation

The main hospital dataset contains 200 rows and 20 original data fields, including patient identifiers, demographic attributes, admission/discharge dates, department, doctor, diagnosis, treatment type, bed status, wait time, length of stay, billing, insurance coverage, satisfaction score, and readmission flag.

Field	Use in Analysis
Patient ID	Patient / admission count
Age, Gender, Ward	Demographic and ward-level filtering
Admission / Discharge Date	Admission trends and time analysis
Department	Department comparison and relationships
Treatment Type	Treatment distribution
Bed Status	Bed occupancy and availability
Wait Time (mins)	Operational delay analysis
Length of Stay (days)	Average stay KPI
Bill Amount	Revenue and billing analysis
Patient Satisfaction (1–10)	Patient satisfaction KPI
Re-admission Flag	Readmission rate analysis

A separate department-level resource dataset supports the Resource Utilization page. It contains Department, Total Staff, Allocated Staff, Total Equipment, and Equipment In Use, and is related to the hospital table through the

Department field. These staff/equipment figures are planning-level sample values, since the original patient-level dataset did not include staff and equipment observations.

Data Cleaning and Transformation

- Reviewed records for completeness and consistency
- Standardized date fields for admission and discharge analysis
- Used consistent department labels for filtering and relationships
- Prepared numeric fields (wait time, length of stay, billing, satisfaction) for aggregation
- Loaded the cleaned hospital and resource datasets into Power BI and created the model relationship on Department

5.3 Data Model

The hospital dataset acts as the main patient-level fact table, and the department-level resource table supports staff and equipment analytics. Department is the common key: the resource table has one row per department and relates to multiple patient records in the hospital table, giving a one-to-many relationship that powers department-based resource analysis.

5.4 Dashboard Planning

Dashboard	Primary Analytical Purpose
Hospital Overview	Overall hospital KPIs, admissions, occupancy, readmissions, monthly operational trends
Patient Flow	Admissions, discharges, patient movement, average stay, peak load
Department Analytics	Patient volume, readmissions, department comparison, efficiency and capacity
Resource Utilization	Beds, staff allocation, equipment utilization, capacity and resource availability

6. KPI Engineering

DAX measures summarize hospital performance and resource utilization throughout the report. The exact measure logic is maintained in the Power BI file (.pbix); the table below documents the analytical meaning of the principal KPIs, with representative DAX patterns.

KPI	Definition / DAX Pattern
Total Patient Admissions	COUNTROWS(Hospital) — count of patient admission records
Hospital Bed Occupancy Rate	DIVIDE(Occupied Beds, Total Beds) — share of tracked beds/records marked occupied
Average Patient Length of Stay	AVERAGE(Hospital[Length of Stay]) — average days between admission and discharge
Patient Readmission Rate	DIVIDE(Readmissions, Total Admissions) — % of records flagged as readmissions
Average Patient Satisfaction Score	AVERAGE(Hospital[Satisfaction Score])
Total Hospital Revenue	SUM(Hospital[Bill Amount])
Department Efficiency Score	Composite department performance measure used for department comparison; formula validated for scale and interpretation
Staff Allocation %	DIVIDE(Allocated Staff, Total Staff)
Equipment Utilization %	DIVIDE(Equipment In Use, Total Equipment)
Available Beds	COUNTROWS(FILTER(Hospital, Hospital[Bed Status]="Available"))

7. Dashboard Implementation

7.1 Hospital Overview & Performance

Provides a consolidated view of overall hospital performance and key operational trends, including total admissions, readmission rate, bed occupancy rate, average patient satisfaction, and total hospital revenue, alongside monthly admissions, length-of-stay, and readmission trends, plus treatment-type and department-level comparisons.

Figure 1: Hospital Overview & Performance dashboard

7.2 Patient Flow & Admission Analysis

Focuses on how patients enter, move through, and leave the hospital — total admissions and discharges, average length of stay, readmission rate, monthly admission trend, admission/discharge comparison, gender distribution, peak patient load, and patient movement by department.

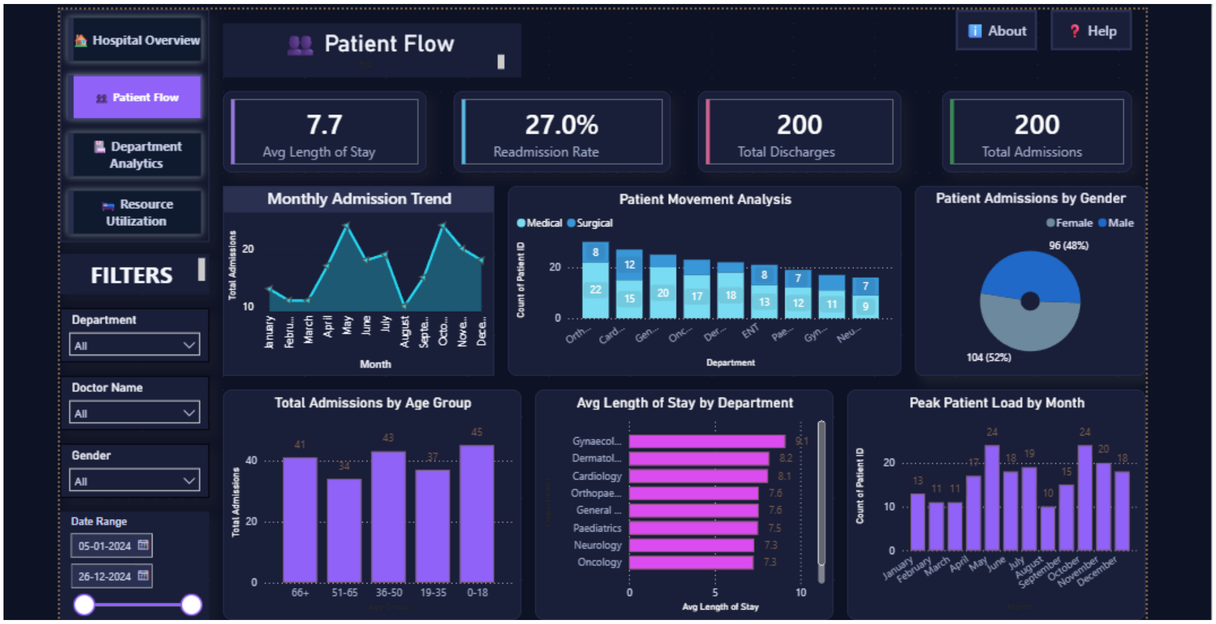

Figure 2: Patient Flow & Admission Analysis dashboard

7.3 Department Performance Analytics

Compares hospital departments using volume, stay, readmission, revenue, and capacity-oriented views, including patient volume by department, average length of stay by department, readmission rate by department, department revenue comparison, treatment capacity/admissions by department, and department-level KPI cards.

Figure 3: Department Performance Analytics dashboard

7.4 Hospital Resource Utilization

Monitors bed utilization and extends the analysis to staff and equipment utilization by department, covering total/available/occupied beds, bed utilization rate, monthly bed utilization trend, staff allocation by department, equipment utilization by department, resource availability, and department capacity overview.

Figure 4: Hospital Resource Utilization dashboard

7.5 Navigation and Interactivity

The final report includes page navigation across Hospital Overview, Patient Flow, Department Analytics, Resource Utilization, About MedTrack DV, and Help. The analytical pages also include slicers and interactive visuals that let users filter and explore the report from multiple operational perspectives.

- Page navigator for movement between dashboard modules
- Slicers for department, date/time, and other available dimensions
- Cross-filtering between charts and KPI cards where applicable
- Action buttons for navigation to About and Help pages
- Consistent dashboard layout to support usability

8. Testing and Validation

Testing focused on validating the correctness of KPI calculations, the behavior of interactive elements, and the consistency of the report's visual presentation.

- Validated all DAX KPI calculations against source data
- Checked slicers, cross-filtering, and page navigation on every dashboard
- Verified admission, discharge, readmission, occupancy, and length-of-stay logic
- Confirmed one-to-many relationship behavior between the resource and hospital tables
- Reviewed titles, units, percentage formatting, data labels, and visual consistency
- Documented identified issues and corrections in a QA checklist

9. Key Findings

The hospital dataset contains 200 patient records distributed across 9 departments. Based on the prepared data, the overall readmission rate is approximately 27.0%, the average length of stay is 7.7 days, and the average patient satisfaction score is 5.4 out of 10. These values provide a high-level baseline for the dashboard and can be explored further by month, department, gender, and treatment type.

Orthopaedics has the highest patient volume in the dataset, with 30 records. Department-level views in the report allow this patient-load pattern to be compared against length of stay, readmissions, billing, and resource availability.

10. Limitations

- The project is based on a limited academic/sample dataset and should not be interpreted as a live hospital information system
- Staff and equipment values were added as a separate planning/sample resource table, since those fields were not present in the original patient dataset
- The dashboard is intended for operational analytics and learning purposes, not clinical diagnosis or medical decision-making
- Composite metrics such as Department Efficiency Score should be interpreted only after validating the exact DAX formula and intended score range

11. Conclusion

MedTrack DV demonstrates an end-to-end hospital analytics workflow using Microsoft Power BI. The solution integrates patient-level operational data and department-level resource data into four connected dashboards that address hospital performance, patient flow, department analytics, and resource utilization.

Through KPI cards, trend visuals, department comparisons, slicers, and navigation, the project provides a clear analytical interface for understanding hospital operations. The completed dashboard suite serves as a portfolio-ready demonstration of data cleaning, data modeling, DAX-based KPI engineering, interactive visualization, and healthcare operations analytics.

12. Future Scope

- Connect to live hospital databases or APIs for near-real-time monitoring
- Add forecasting for patient admissions, bed demand, and staffing requirements
- Develop alerts for high occupancy, long wait times, and abnormal readmission patterns
- Add drill-through patient/department detail pages and richer tooltips
- Publish the report through Power BI Service with role-based access and scheduled refresh
- Replace sample resource values with verified staff, equipment, and shift-level operational data